“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 14

Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!

- Quiz on Friday on polar coordinates and beginning of triple integrals

- Exam 1 – One-Page Correction (+5) – due 3/9 – can be found on HuskyCT > Quizzes and Exams (will not be accepted late)

ALVARO: Start the recording!

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More on Triple Integrals in Cylindrical Coordinates

Today – Triple Integrals in Cylindrical Coordinates

- Cylindrical Coordinates
- Triple Integrals in Cylindrical Coordinates

Cylindrical Coordinates

$$x = r \cos\theta \qquad y = r \sin\theta \qquad z = z$$

$$r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x} \qquad z = z$$

Example: Evaluate the value of the triple integral

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2} (x^2+y^2)\ dz\,dy\,dx$$

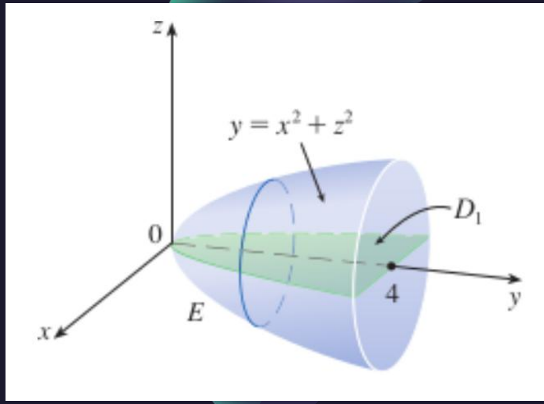
Example: Evaluate the value of the triple integral

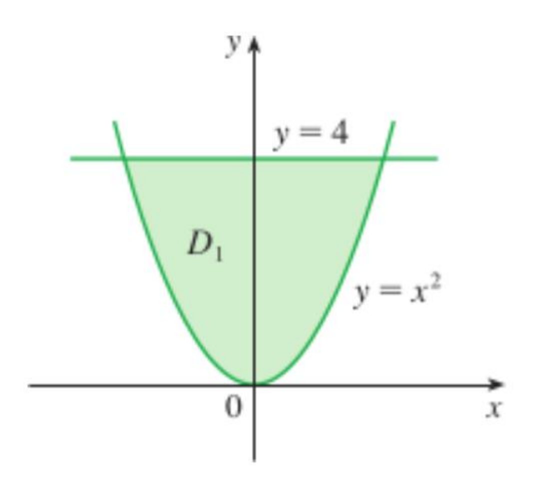
$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2} (x^2+y^2)\ dz\, dy\, dx$$

Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2+z^2}\,dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.





Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2+z^2}\, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane z = 4, and above $z = 1 - x^2 - y^2$.

Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane z = 4, and above $z = 1 - x^2 - y^2$.

Questions?

ALVARO: Start the recording!

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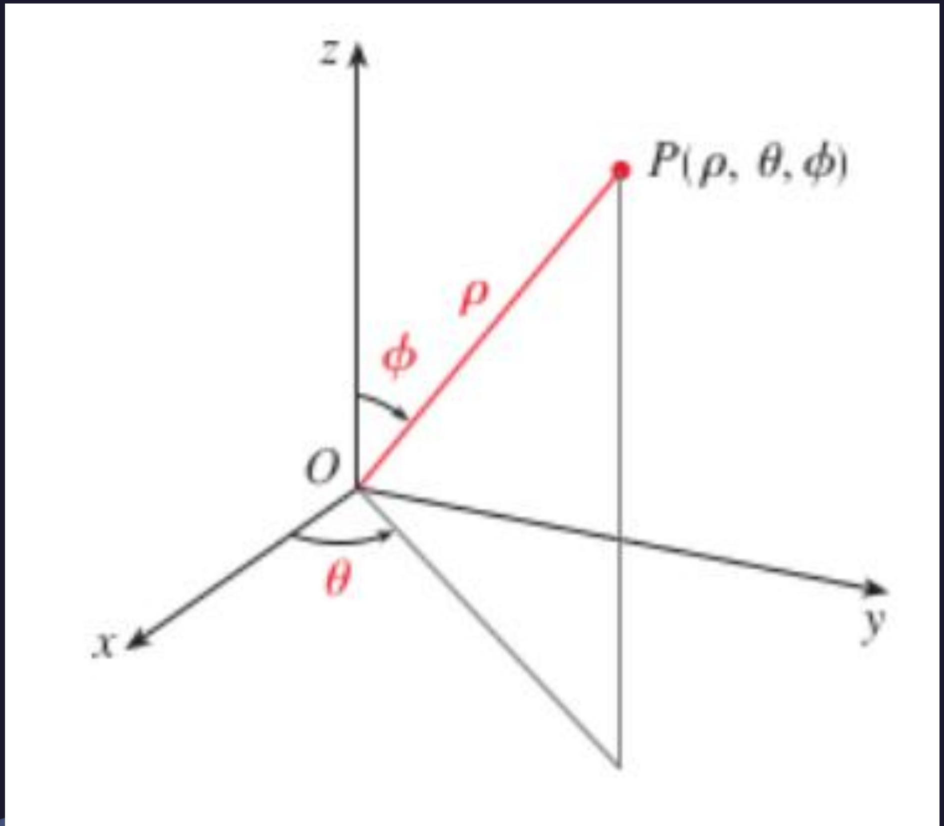
Triple Integrals in Spherical Coordinates

Today – Triple Integrals in Spherical Coordinates

- Spherical Coordinates
- Triple Integrals in Spherical Coordinates

Spherical Coordinates

Spherical Coordinates



$$x = \rho \sin \phi \cos \theta \qquad y = \rho \sin \phi \sin \theta \qquad z = \rho \cos \phi$$

$$\rho^2 = x^2 + y^2 + z^2$$

Cylindrical vs Spherical Coordinates

$$x = r\cos\theta \qquad y = r\sin\theta \qquad z = z$$

$$x = \rho\sin\phi\cos\theta \qquad y = \rho\sin\phi\sin\theta \qquad z = \rho\cos\phi$$

$$r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x} \qquad z = z$$

$$\rho^2 = x^2 + y^2 + z^2$$

Example: Sketch the surface given by $\rho = k$.

Example: Sketch the surface given by $\theta = k$.

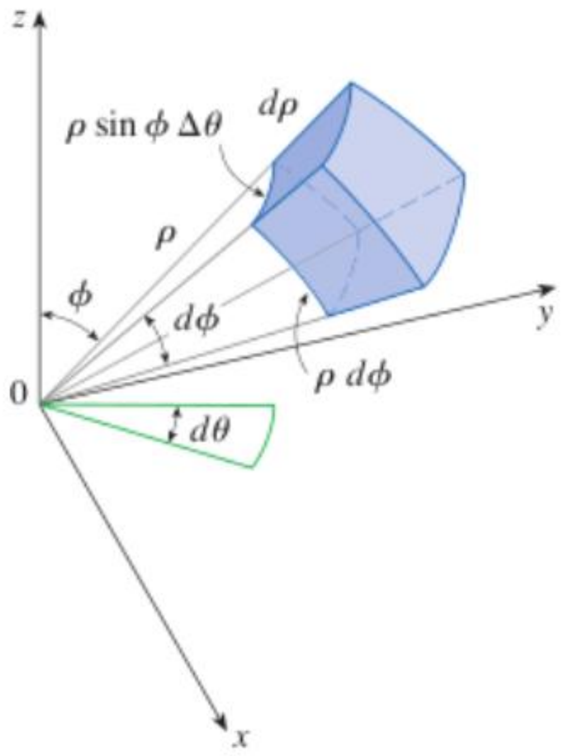
Example: Sketch the surface given by $\phi = k$.

Example: Sketch the “spherical wedge” given by

$$a \leq \rho \leq b\,, \alpha \leq \theta \leq \beta,\ c \leq \phi \leq d.$$

Triple Integrals in Spherical Coordinates

Volume element in spherical coordinates: $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$



Triple Integrals in Spherical Coordinates

$$\iiint\limits_E f(x, y, z)\, dV$$

$$= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid \ a \leqslant \rho \leqslant b, \alpha \leqslant \theta \leqslant \beta, c \leqslant \phi \leqslant d\}$$

Example: Use triple integrals to find the volume of the sphere of radius R.

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Example: Evaluate

$$\iiint_E \; e^{(x^2+y^2+z^2)^{\wedge}3/2} \, dV$$

where E is the unit sphere/ball.

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$$\iiint_E e^{(x^2+y^2+z^2)^\wedge 3/2}\, dV$$

where E is the unit sphere/ball.

Example: Use triple integrals to find the volume of the "ice-cream cone" that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $z = x^2 + y^2 + z^2$.

[Extra]

Example: Use triple integrals to find the volume of the "ice-cream cone" that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $z = x^2 + y^2 + z^2$.

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Questions?

Thank you

Until next time.